Sensor report

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***Abstract*—This report compiles the different types of sensors, covering inertial measurement units (IMUs), GNSS receivers, wheel speed sensors, mono and stereo cameras, LiDAR and the supporting chassis sensors. For each sensor we explain the physics behind the measurement, the usecase in thecar, , the cost and where to buy in India and how long it takes.**

Keywords—Formula Student, IMU, GNSS, RTK, wheel speed sensor, stereo camera, LiDAR, sensor fusion, sourcing in India

I. Introduction

This report covers the sensors named in the task brief (IMU, GPS/GNSS, wheel speed sensors, mono and stereo cameras, LiDAR) and, for completeness, the chassis and powertrain sensors every FS car carries. The report also has vocabulary used to compare sensors.Main sensor in the order: working principle, key specifications, use case for us, candidate parts, and cost with sourcing. It also summarises the remaining sensors, and discusses integration and gives a consolidated cost and sourcing list maiiinly in india.

II. Sensor Selection Metrics

- Range and resolution: the span of the measured quantity and the smallest change that can be reported.
- Accuracy, bias and noise: systematic offset (bias) scatter (noise density) and how accuracy, bias and noise vary with temperature and time. For sensors the Allan variance is the standard tool to separate accuracy, bias and noise .Bandwidth and update rate: a sensor must be sampled well above the dynamics of interest. Chassis motion is below about 20 Hz, but damper and suspension events and engine vibration reach hundreds of Hz.
- Latency and synchronisation: a 10 ms delay at 20 m/s moves the car 0.200 m, so every sensor needs a trustworthy timestamp.
- Interface: analogue voltage, I2C/SPI, UART, CAN at 1 Mbit/s, USB 3, or Ethernet for high-bandwidth devices such as LiDAR.
- Environment: vibration, temperature, water and dust ingress, electromagnetic noise from the ignition system, motor controller or tractive system, and sunlight for optical devices.
- Cost and availability: total landed cost in rupees including GST and import duty, lead time, warranty and counterfeit risk.

III. Inertial Measurement Units

A. Working Principle

A MEMS IMU combines a three-axis accelerometer and a three-axis gyroscope, often with a three-axis magnetometer to form a 9-axis device. A MEMS accelerometer contains a silicon proof mass suspended on springs. Under acceleration the mass moves relative to the frame and the change in capacitance between interleaved comb fingers is converted to a digital word. It measures specific force, which means gravity is always present in the reading when the sensor is at rest.

A MEMS gyroscope drives a small mass into vibration at a known velocity.The magnetometer (Hall or anisotropic magnetoresistive) senses the Earth's field to give an absolute heading, but it is easily corrupted by hard-iron and soft-iron distortion from motors, steel tubes and battery currents, so on a racecar its output is normally treated with suspicion.

B. Error Sources and Why Fusion Is Needed

Errors are scale-factor error, axis misalignment, temperature drift, cross-axis sensitivity and vibration rectification, which is a particular problem on a single-cylinder FS engine.

C. Use Cases onthe Car

- Vehicle dynamics: lateral and longitudinal acceleration (g-g diagram), yaw rate and roll and pitch for tyre and suspension analysis [5].
- State estimation: yaw rate and acceleration drive the EKF that estimates velocity and side-slip angle between GNSS fixes or wheel-speed updates.
- Driverless: ego-motion compensation of LiDAR point clouds and dead-reckoning when vision is poor [6].
- Traction and launch control: acceleration is used to cross-check wheel speed during wheel spin.

D. Candidate Parts, Cost and Sourcing

1) Entry level, MPU-6050 (GY-521 module): A 6-axis device with 16-bit converters, selectable ranges of ±2 to ±16 g and ±250 to ±2000 °/s, and an I2C interface [7]. It is widely stocked in India at ₹199.00 including GST (Probots) to ₹274.94 including GST (Zbotic), and marketplace sellers list it for roughly ₹100 to ₹250. It is excellent for learning and for bench tests, but it is an ageing part, has relatively high noise, is sold in many counterfeit and clone versions, and has no magnetometer. The GY-87 board that adds a magnetometer and barometer was listed at ₹949.00 including GST.

2) Mid level, Bosch BNO055: A 9-axis sensor with an on-board processor that outputs Euler angles and quaternions at 100 Hz [8]. The Adafruit breakout was listed by Evelta (Mumbai warehouse) at ₹3,518.00 excluding GST, which is ₹4,151.24 after 18 % GST. Its convenience is balanced by a closed fusion algorithm that is hard to tune and is sensitive to magnetic disturbances.

IV. GPS and GNSS Receivers

A. Working Principle

A Global Navigation Satellite System receiver measures the travel time of signals broadcast by satellites.

A standalone receiver achieves metre-level accuracy; the ZED-F9P quotes better than 1.500 m in standalone mode [9].

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Real-time kinematic (RTK) positioning improves this to centimetres. A fixed base station or an NTRIP network supplies RTCM correction data, the rover resolves the integer number of carrier-phase cycles to each satellite, and the position error falls to about 0.010 m plus 1 ppm of the baseline. On a 10 km baseline this is 0.020 m [9]. Multi-band (L1/L2) receivers resolve ambiguities in seconds and are far more robust than single-band units. Two receivers in a moving-base arrangement also give true heading from the vector between two antennas, which does not depend on motion or magnetometers.

Constellations include GPS, GLONASS, Galileo and BeiDou. India operates NavIC (IRNSS), a regional system covering India and its surroundings, and GAGAN, an augmentation service. Most low-cost multi-band modules, including the ZED-F9P, list GPS, GLONASS, Galileo, BeiDou, QZSS and SBAS but not NavIC [9], so NavIC cannot be assumed for such parts.

B. Key Limitations

- Update rate: even at 20 Hz the car travels 1.000 m between fixes at 20 m/s, whereas a 100 Hz IMU reduces the gap to 0.200 m. This is the main reason GNSS is fused with an IMU.
- Sky view and multipath: buildings, grandstands and tree cover degrade the solution, and low-cost patch antennas pick up reflections from the car's own bodywork. A ground plane and antenna placement high on the main hoop matter.
- RTK needs a correction link: a base station with a radio or an NTRIP internet connection must be planned for the test site.
- Rules: competitions differ in what the localisation pipeline may use. Teams commonly use GNSS as ground truth for validating SLAM and speed estimation, but this must be checked against the current rules [1].

C. Use Cases onthe Car

- Helps to time the laps, track mapping and making sure no crahses happen and racing-line comparison from logged position.
- Ground-truth speed to calibrate wheel speed sensors and to measure slip.
- Driverless: ground truth for evaluating SLAM, and global-frame velocity for state estimation [6].

D. Candidate Parts, Cost and Sourcing

Basic single-band modules in the NEO-M8 class are indicatively priced at a few hundred to a couple of thousand rupees and are sufficient for logging only. For RTK the practical choice is a u-blox ZED-F9P board. GenX Battery sells Indian-assembled ZED-F9P receivers: the lightweight F9P Air (20 g, helical antenna and compass) at 15,392.00 excluding GST, and the F9P Tough (aluminium case, external antenna, USB and two serial ports) at 15,688.00 excluding GST, which is 18,511.84 after 18 % GST. ArduSimple (Spain) ships worldwide with a two-year warranty: simpleRTK2B Budget boards from €176, the Lite at €189 and a starter kit with an IP67 ANN-MB multiband antenna at €211, before shipping, GST and duty. Expect to add a second board as a base station, or use a public NTRIP service where available.

V. Wheel Speed Sensors

A. Working Principle

A wheel speed sensor generates a pulse train as a toothed ring (tone wheel) or a set of magnets rotates past it. Three technologies are common. A variable-reluctance (VR) or inductive pickup generates an AC voltage whose amplitude and frequency grow with speed; it is simple and rugged but needs signal conditioning and gives a weak signal at low speed. A Hall-effect sensor is active and outputs a clean digital waveform down to zero speed, which is why modern ABS systems use it. Optical or magnetic encoders give much finer angular resolution but are less tolerant of dirt and heat.

Use Cases on the Car

. Vehicle speed estimation for the EKF, and odometry for driverless SLAM.

B. Candidate Parts, Cost and Sourcing

The most economical option for learning is the A3144E Hall switch, listed at ₹59.00 including GST by Probots (₹50 plus GST), and it needs a magnet ring on the wheel. Inductive barrel sensors such as the LJ12A3-4-Z/BX (NPN, 6 to 36 V, 4 mm nominal sensing distance) are listed in India at about ₹159 to ₹412, for example ₹294 including GST for the M12 variant at Probots, but their switching frequency is limited, so their datasheet must be checked against the highest wheel frequency. For the car itself, automotive active ABS-type Hall sensors (from Bosch, Honeywell, Allegro or Melexis ICs on a custom bracket) are recommended. They are not catalogue items on the hobby shops, so they are sourced through automotive spares dealers or from Mouser India, DigiKey India or Element14 India, and a realistic estimate is a few thousand rupees per set of four.

VI. Cameras: Monocular and Stereo

A. Working Principle

A camera projects the 3-D world onto a 2-D sensor. Under the pinhole model a point at (X, Y, Z) maps to pixel coordinates

Some intrinsic parameters and the lens distortion coefficients are found by calibration, and extrinsic calibration gives the pose of the camera relative to the car [10]. Important sensor properties are resolution, frame rate, dynamic range, field of view and shutter type. A global shutter exposes all pixels simultaneously, whereas a rolling shutter exposes row by row and skews fast-moving scenes; on a car moving at 20 m/s a global shutter is strongly preferred.

B. Monocular Cameras

A single camera gives rich semantic information (cone colour, signs, track edges) but no direct depth. Depth can be inferred when the object size is known].

C. Stereo Cameras

Two cameras separated by a baseline B observe the same scene are stereo cameras basically.

D. Use Cases onthe Car

- Driverless cone detection, colour classification and, with a LiDAR, sensor fusion of semantics and geometry [6].
- Visual odometry and SLAM, with an IMU for scale and robustness.

- Driver-facing or on-board video synchronised with logged data for post-run analysis.

E. Candidate Parts, Cost and Sourcing

Entry-level USB webcams or Raspberry Pi cameras are adequate for developing the software pipeline, but they use rolling shutters. For the car, global-shutter machine-vision cameras (Basler, FLIR, Allied Vision, or global-shutter OV9281-based modules) with a fixed lens are preferred. For stereo, Intel RealSense D435 and D435i cameras (wide field of view, global-shutter depth sensors, 50 mm baseline) are available in India through ThinkRobotics, where the D435/D435i/D455 family is listed in the range of roughly ₹35,000 to ₹60,000 (the D435i at about ₹43,999.99 on sale). Stereolabs ZED 2i (120 mm baseline) and Luxonis OAK-D Lite cameras are imported through distributors or directly; as a rough estimate (to be re-verified) they cost about US$450 to 550 and US$150 to 200 respectively before shipping, GST and customs duty. Cameras also require a compute unit (an NVIDIA Jetson-class module or a laptop-class computer), which is often more expensive than the camera itself.

VII. LiDAR

A. Working Principle

LiDAR (Light Detection and Ranging) measures distance by timing a laser pulse. In the time-of-flight (ToF) method the range is

$$d = c\,\Delta t / 2 \tag{10}$$

so an object at 20 m returns the pulse after 133.426 ns, and a timing resolution of 1 ns corresponds to a range resolution of 0.150 m. Some sensors use amplitude-modulated continuous wave (phase shift) or frequency-modulated continuous wave (FMCW) ranging, and low-cost 2-D scanners use laser triangulation, in which the position of the returning spot on a sensor line gives the range. A spinning multi-beam unit stacks 16 to 128 laser channels vertically and rotates them to produce a point cloud of hundreds of thousands of points per second, while solid-state and non-repetitive-scan units (such as Livox) trade field of view for lower cost and no moving parts. Each point also carries a reflectivity (intensity) value, which helps to separate cones from the ground and their colour bands.

B. Processing Pipeline

A typical cone-detection pipeline first removes the ground plane (RANSAC or grid-based methods), then clusters the remaining points (Euclidean clustering or DBSCAN), filters clusters by size and shape, and finally classifies them, optionally by projecting the clusters into the camera image for colour [6]. Motion distortion is corrected using the IMU, and the LiDAR must be accurately extrinsically calibrated to the camera and the vehicle frame.

C. Advantages and Limitations

- Direct, accurate geometry: centimetre-level ranging with little dependence on lighting.
- Sparse at distance: beams diverge, so a small cone returns only a few points beyond about 20 m for a 16-channel unit; higher channel counts help.
- No colour: colour classification of cones needs a camera or intensity patterns.
- Environment: heavy rain, dust and direct sunlight can degrade returns, and low-cost triangulation scanners are designed for indoor use or shade only.
- Cost and power: the most expensive sensor on the car, with a power draw of several watts.

D. Use Cases on this Car

- Driverless cone detection and mapping; the primary perception sensor for most top teams .
- SLAM and localisation against the cone map, together with IMU and wheel speed.
- Obstacle detection for safety-critical stopping decisions.

E. Candidate Parts, Cost and Sourcing

1) Entry level: Slamtec RPLIDAR A1/A2: A 2-D, 360° triangulation scanner with a 12 m range (A1M12), 5.5 Hz default and 10 Hz maximum scan rate and about 1450 samples per revolution. ThinkRobotics lists the A1M12 at ₹10,699.99 (MRP ₹12,499.99). It is useful for learning ROS 2 SLAM and for indoor tests, but it works well only indoors or in shade and has a single scan plane, so it is not suited to cone detection on a sunny track.

2) Competition grade: Multi-channel spinning LiDARs (Velodyne VLP-16 and its successors, Ouster OS1, Hesai Pandar, RoboSense) or Livox units. As rough estimates to be re-verified, these cost from about US$750 to 1,000 for a Livox Mid-360 class unit to several thousand US dollars for 16- to 64-channel spinning units, plus import GST. There is no catalogue price in India, so a quotation from the manufacturer's authorised distributor is requested; sponsorship or second-hand units from robotics labs are the usual routes for student teams.

VIII. Other Chassis and Powertrain Sensors

Several simpler sensors complete the data-acquisition and safety system. The list below gives their physical principle and typical use. The prices are indicative estimates for common catalogue parts, not quoted listings, and must be re-verified before purchase.

- Steering angle: magnetic rotary encoder (AS5600 or AS5048) or potentiometer for driver input and driverless steering feedback, about ₹300 to 1,500.
- Throttle/APPS: two independent Hall or potentiometer channels with a plausibility check, as required by the rules [1], [2], about ₹500 to 2,000.
- Brake pressure: strain-gauge pressure transducer (0.5 to 4.5 V) for brake bias and brake plausibility, about ₹2,000 to 6,000.
- Damper travel: linear potentiometer or LVDT for suspension tuning and ride height, about ₹1,500 to 5,000.
- Tyre temperature: infrared thermopile (MLX90614 class) for tyre-window analysis, about ₹800 to 2,500.
- Temperatures: NTC thermistor or thermocouple for coolant, oil, motor and battery, about ₹50 to 500.

- Current and voltage: Hall-effect current sensor on the EV tractive system for energy and efficiency, about ₹800 to 5,000.

IX. Integration and Sensor Fusion

A. Mounting and Electrical Practice

- Mount the IMU rigidly near the centre of gravity on the chassis, aligned with the vehicle axes, and isolate it from engine vibration if its noise floor is sensitive.
- Place the GNSS antenna on a ground plane with a clear view of the sky, away from cameras and switching power supplies.
- Use shielded twisted pairs, star grounding and ferrite beads, and keep sensor wiring apart from ignition coils and motor phase cables.
- Power sensitive sensors from a regulated, filtered supply; LiDAR and compute modules need adequate current and fusing.

X. Consolidated Cost and Sourcing in India

The prices below were read from Indian retailer listings during the research for this report (October 2026) and are shown as listed; they change often and must be re-checked before an order. The ZED-F9P receivers by GenX Battery are listed without a currency symbol in the captured text and are assumed to be in INR. ArduSimple prices are in euros and exclude shipping, GST and customs duty.

- GY-521 MPU-6050: ₹199.00 to ₹274.94 including GST (Probots, Zbotic).
- GY-87 10-DOF IMU: ₹949.00 including GST (Probots).

Buying advice for India follows from these listings. Hobby modules are cheap and available within one to three days from Bengaluru, Delhi and Mumbai suppliers such as Probots, Robu, ThinkRobotics, Evelta and Zbotic, but quality varies and clones are common, so buy critical chips from authorised distributors (Mouser India, DigiKey India, Element14 India, RS Components) and request a GST invoice, because it supports the team's expense records. Imported competition-grade parts attract basic customs duty and 18 % IGST and take weeks, so orders should be placed early, and manufacturers frequently offer academic discounts or sponsorship to Formula Student teams.

A staged purchase strategy keeps the budget under control. Stage 1 (bench and learning) uses a GY-521, a Hall sensor and a low-cost LiDAR or webcam, costing a few thousand rupees in total. Stage 2 (data-acquisition car) adds a BNO055 or a better IMU, a ZED-F9P receiver and the chassis sensors from Section VIII. Stage 3 (driverless) adds a global-shutter or stereo camera, a multi-channel LiDAR and the compute platform, which dominates the total cost.

XI. Conclusion

A driverless car needs sensors at three levels: rules-driven safety sensors, data-acquisition sensors for the feedback and perception sensors for autonomy. IMUs, GNSS receivers and wheel speed sensors form the core of state estimation, while cameras and LiDAR provide perception. In India, low-cost modules are readily available within like some days, whereas RTK receivers are available locally from Indian assemblers, and cameras and LiDARs mostly require imports, early ordering and sponsorship.

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